# Athena Autocallable Notes on the CAC40 under a Calibrated Heston Model: Fair Coupons, Scenarios and Greeks

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**Abstract**

We price single-underlying Athena autocallable notes, whose coupon is paid at the call only, on the CAC40 of 12 February 2025. The underlying follows a Heston model calibrated to the implied volatility surface (RMSE 0.42 volatility point), with a drift read from a forward curve bootstrapped from put–call parity. The note is described by a single parametrisation covering the observation frequency, snowball or fixed coupons, autocall barrier and step-down, non-call period, European, American continuous or American daily capital protection, airbag, buffer, capital guarantee, range accrual coupon, duo call mechanism and look back. The notes are priced by Monte Carlo with antithetic variates and a control variate; the American protection is handled with the Brownian bridge crossing probability as a conditional expectation. The pricer recovers three closed-form limit cases within one standard error. The base Athena (4 years, quarterly observations, snowball coupon, autocall at 100%, European protection at 60%) has a fair coupon of 10.54% a year: it is called at the first date with probability 55.4%, lives 1.03 years on average and loses capital with probability 5.2%. We report its sensitivities to the barriers and to the market (volatility, dividends, rates, issuer spread, maturity, skew), its hedge, and the fair coupon of 18 variants, from 4.83% for a full capital guarantee to 12.53% for an American protection. The note is short volatility ($-0.32$ per vol point on a nominal of 100) and short dividends ($-0.69$ per point of carry).

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# 1 Introduction

In an AutoCallable note, the investor receives coupons well above the risk-free rate, and the note is redeemed early ("called") as soon as the index is above a reference level on an observation date. The coupon is financed by selling a down-and-in put. If the note is never called and the index ends below a protection barrier, the investor bears the loss of the index.

This report studies the **Athena**, whose coupon is paid only when the note is called, usually accumulated over time (*snowball*). The other main family, the Phoenix, whose coupon is paid at each observation date where the index is above a coupon barrier, is treated in a companion report [2]. On top of this structure, issuers combine many features: observation frequency, step-down of the autocall barrier, non-call period, European or American protection, airbag, buffer, capital guarantee, range accrual coupon, duo call mechanism, look back. We will have a look on how to price these different versions in a single framework, on real market data, and measure how each feature moves the fair coupon, the probabilities of the scenarios and the Greeks.

The model is the Heston model calibrated in [1] to the CAC40 implied volatility surface, and the Monte Carlo engine is the one used for barrier options in [3]. The scripts and notebooks are open source; Section 6 describes the code.

# 2 Products

## 2.1 Common structure

A note of nominal $N$ on an index $S$ has an initial fixing $S_{\text{ref}}$ and observation dates $0 < t_1 < \cdots < t_n = T$. The performance at $t_k$ is $P_k = S_{t_k}/S_{\text{ref}}$. All barriers are fractions of $S_{\text{ref}}$: autocall barriers $B_k^{\text{call}}$ and protection barrier $B^{\text{prot}}$. The dates $t_k$ with $t_k \geq t_{\text{nc}}$ are callable, $t_{\text{nc}}$ being the end of the non-call period; the maturity is always callable. The first callable date where $P_k \geq B_k^{\text{call}}$ is the call date $\tau$.

**Redemption at maturity.** If the note is never called, it is redeemed at $T$ at

$$R_T = N\left[(1 - \mathbf{1}_{\text{KI}}) + \mathbf{1}_{\text{KI}} \max\Big(\phi,\ \min\left(1,\ P_T/K_{\text{put}} + b\right)\Big)\right], \tag{1}$$

where $\mathbf{1}_{\text{KI}}$ is the knock-in event of the protection, $K_{\text{put}}$ the strike of the short put $\phi$ the capital floor and $b$ the buffer ($b = 0$ by default). The knock-in event is

- *European protection*: $P_T < B^{\text{prot}}$;
- *American protection*: the index touches $B^{\text{prot}} S_{\text{ref}}$ at any time before maturity, continuously or on daily closes.

The standard note has $K_{\text{put}} = 1$ and $\phi = 0$: below the barrier, the investor receives $NP_T$ and the redemption jumps from $N$ to $NB^{\text{prot}}$ at the barrier. The *airbag* note has $K_{\text{put}} = B^{\text{prot}}$: the redemption is continuous at the barrier and equal to $NP_T/B^{\text{prot}}$ below it. A *capital guarantee* sets $\phi > 0$. A *buffer* sets $b > 0$ with $B^{\text{prot}} = 1 - b$: the investor only loses below $1 - b$, one for one, without loss jump. Figure 1 draws the three profiles.

## 2.2 Athena

The Athena pays a coupon only with the redemption at the call date. If $\tau = t_k$, the investor receives

$$N(1 + c_k), \qquad c_k = \begin{cases} c\, t_k & \text{snowball (the coupons accumulate),} \\ c\,(t_k - t_{k-1}) & \text{fixed coupon,} \end{cases} \tag{2}$$

with $c$ the coupon rate per annum. If the note is not called, no coupon is paid and the redemption is (1). A call at maturity ($P_T \geq B_n^{\text{call}}$) pays the last coupon.

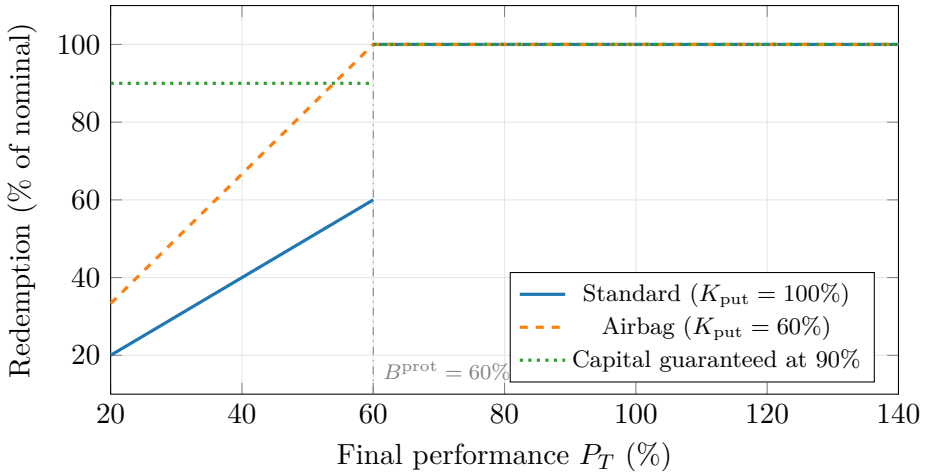



Figure 1: Redemption at maturity of a note that was never called, European protection at 60%.

**Optional features.** Three features change the call or the coupon:

- *range accrual*: the coupon only accrues on the dates where $L \leq S_t/S_{\text{ref}} \leq U$, so that a call at $t_k$ pays $c$ times the time spent in the range instead of $c\,t_k$;
- *duo*: a second call level $B_k^{\text{duo}} < B_k^{\text{call}}$; if $B_k^{\text{duo}} \leq P_k < B_k^{\text{call}}$, the note is called with a fraction $\alpha$ of the coupon;
- *look back*: $S_{\text{ref}}$ is the minimum (or the average) of the index over $[0, t_{\text{lb}}]$, path by path, instead of the initial fixing, all the barriers being fractions of this reference.

## 2.3 Parametrisation

Table 1 lists the parameters of the class `Athena`; every variant of Section 5 is one of these parameters changed.

| Feature | Parameter | Values |
|---|---|---|
| Maturity, initial fixing, nominal | `maturity`, `initial_fixing`, `nominal` | years, index level, 100 |
| Observation dates | `frequency` or `observation_dates` | monthly to annual, or any list ending at $T$ |
| Coupon rate | `coupon_rate` | per annum, fraction of the nominal |
| Athena coupon | `coupon_type` | `snowball`, `fixed` |
| Autocall barrier | `autocall_barrier`, `step_down` | fraction of $S_{\text{ref}}$, decrease per observation |
| Non-call period | `first_call_date` | years |
| Capital protection | `protection_barrier`, `protection_type` | fraction of $S_{\text{ref}}$; `european`, `american`, `american_discrete` |
| Airbag, guarantee | `put_strike`, `floor` | fraction of $S_{\text{ref}}$, fraction of the nominal |
| Buffer | `buffer` | fraction of $S_{\text{ref}}$, with `protection_barrier` $= 1 - b$ |
| Range accrual | `accrual_range` | $(L, U)$, fractions of $S_{\text{ref}}$ |
| Duo | `duo_barrier`, `duo_coupon_ratio` | fraction of $S_{\text{ref}}$, fraction $\alpha$ of the coupon |
| Look back | `lookback`, `lookback_type` | years, `min` or `average` |

Table 1: Parameters of the Athena (`Products/Athena/athena.py`).

## 3 Market Data and Model

The data are the CAC40 option chain of 12 February 2025 (spot $S_0 = 8042.19$, 142 quotes on 13 expiries from 9 days to 4.86 years) and the EURIBOR 6M zero-coupon curve. The forward curve $F(0,t)$ is bootstrapped from put–call parity with a piecewise constant borrow-plus-dividend yield $q(t)$ [1]; its integral reaches 0.136 at 4.86 years, so dividends lower the 4-year forward by about 11% with respect to a zero carry. Under $\mathbb{Q}$,

$$\frac{\mathrm{d}S_t}{S_t} = \partial_t \ln F(0,t)\,\mathrm{d}t + \sqrt{v_t}\,\mathrm{d}W_t^S, \qquad \mathrm{d}v_t = \kappa(\theta - v_t)\,\mathrm{d}t + \xi\sqrt{v_t}\,\mathrm{d}W_t^v, \qquad \mathrm{d}\langle W^S, W^v\rangle_t = \rho\,\mathrm{d}t, \tag{3}$$

with the parameters of Table 2. The notes are struck at the money: $S_{\mathrm{ref}} = S_0 = 8042.19$, and $S_{\mathrm{ref}}$ stays fixed when the Greeks bump the market. All the dates of the 4-year notes lie before the last listed expiry (4.86 years).

| $v_0$ | $\theta$ | $\kappa$ | $\xi$ | $\rho$ | IV RMSE |
|---|---|---|---|---|---|
| 0.0197 | 0.0438 | 1.308 | 0.482 | $-0.605$ | 0.421 vol pts |

Table 2: Heston parameters calibrated on the CAC40 surface of 12 February 2025 [1].

# 4 Pricing Method

## 4.1 Simulation

The paths of $(S, v)$ are simulated with an Euler full truncation scheme on a weekly grid (52 steps a year, 208 steps for 4 years), the drift over each step being $\ln(F(0,t_{i+1})/F(0,t_i))$, so that $\mathbb{E}[S_t] = F(0,t)$ at every date. $N = 50{,}000$ paths with antithetic variates are used (seed 42); prices and standard errors are computed on the 25,000 independent pair averages. The observation dates are rounded to the grid; with weekly steps, every monthly, quarterly, semi-annual and annual date falls within half a week of its contractual date.

## 4.2 Cash flows and protection

For each path, the note is followed through its observation dates, each cash flow being discounted with $D(0,t_k) = e^{-r(t_k)t_k}$. The knock-in indicator of (1) is

- *European*: $\mathbf{1}_{\{P_T < B^{\text{prot}}\}}$;
- *American continuous*: replaced by its conditional probability given the simulated path, $1 - \prod_i (1 - p_i)$, where $p_i = \exp(-2\ln(S_{t_i}/H)\ln(S_{t_{i+1}}/H)/(v_{t_i}\Delta t))$ is the probability that the Brownian bridge of $\ln S$ touches $H = B^{\text{prot}} S_{\text{ref}}$ between two grid dates [8];
- *American daily*: the barrier is checked on a daily grid (252 steps a year).

## 4.3 Variance reduction

Besides antithetic variates, the discounted European put struck at $B^{\text{prot}} S_{\text{ref}}$, whose Heston price is known semi-analytically (Lewis formula), is used as a control variate with an optimal coefficient estimated on the paths.

## 4.4 Fair coupon and scenario statistics

On a fixed set of paths, the call dates do not depend on the coupon, and every cash flow is affine in the coupon rate $c$: the price is $V(c) = V(0) + c\,(V(1) - V(0))$, and the coupon that prices the note at par is

$$c^\star = \frac{N - V(0)}{V(1) - V(0)}, \tag{4}$$

obtained with two valuations. It excludes the issuer margin and funding spread. The same paths give the probability of a call at each date, the probability to reach maturity, the probability of a capital loss ($\mathbf{1}_{\text{KI}} = 1$ and $P_T < K_{\text{put}}$), the expected loss at maturity, the expected life $\mathbb{E}[\min(\tau, T)]$ and the expected coupon paid.

## 4.5 Greeks

The Greeks are central finite differences with common random numbers [7], on a nominal of 100: Delta and Gamma (spot $\pm 1\%$, carry kept), Vega and Volga ($\sqrt{v_0}$ and $\sqrt{\theta}$ shifted by $\pm 1$ vol point), Vanna, Rho ($\pm 10$bp on the zero curve, per $+1\%$), Dividend ($\pm 10$bp on $q(t)$, per $+1\%$) and Theta (dates moved by one day). Delta is given per index point: multiply by $S_0/100 = 80.4$ for the value change of a 1% move. Each Greek is reported with its Monte Carlo standard error.

## 4.6 Validation

Three limit cases have a closed form (Table 3). In a flat Black–Scholes market ($S_0 = S_{\text{ref}} = 100$, $r = 3\%$, $q = 1\%$, $\sigma = 20\%$, $T = 2$, $N = 100$, Heston with $v_0 = \theta = \sigma^2$ and $\xi = 10^{-4}$, $10^5$ paths):

- a note never called (autocall barrier out of reach) with a European protection at $B = 60$ is worth $ND(T) - \frac{N}{S_{\text{ref}}}\left[\text{Put}(B) + (S_{\text{ref}} - B)\,\text{DigitalPut}(B)\right]$;
- with an American protection, $ND(T) - \frac{N}{S_{\text{ref}}}\,\text{PDI}(K = S_{\text{ref}}, H = B)$ (Reiner–Rubinstein);
- a note called for sure at $t_1$ is worth $N(1 + c\,t_1)D(t_1)$.

| **Case** | **Reference** | **Monte Carlo** | **Std error** | **Error / std error** |
|---|---|---|---|---|
| Athena never called, European protection (flat) | 92.6369 | 92.6480 | 0.0141 | 0.79 |
| Athena never called, American protection (flat) | 91.5500 | 91.5483 | 0.0220 | −0.07 |
| Athena called for sure at $t_1 = 0.5$, $c = 8\%$ (flat) | 102.4516 | 102.4516 | 0 | exact |

Table 3: Validation of the pricer.

# 5 Athena : Base case

Let us consider, under the risk-neutral measure $\mathbb{Q}$, the general case of an Athena in the base case. To lighten notation, we denote by

$$E_i = \bigcap_{j=1}^{i-1} \{X_j < B_{ac}\}, \qquad i = 1, \dots, N, \qquad E_1 = \Omega, \tag{5}$$

the event that the product has not been called before date $t_i$. The event $\{\tau = t_i\}$ of an early redemption at date $t_i < T$ is then $E_i \cap \{X_i \geq B_{ac}\}$, and $\{\tau = T\} = E_N$.

## 5.1 Price of the Athena

For a coupon $c$, the price of the Athena at inception is

$$P_0 = \mathbb{E}^{\mathbb{Q}}\big[D(\tau)\,\Pi_\tau\big], \tag{6}$$

which, writing the autocall events explicitly, expands as

$$\begin{aligned} P_0 = \sum_{i=1}^{N-1} & D(t_i)\,(1+ic)\,\mathbb{Q}(E_i \cap \{X_i \geq B_{ac}\}) \\ & + D(T)\,(1+Nc)\,\mathbb{Q}(E_N \cap \{X_N \geq B_{ac}\}) \\ & + D(T)\,\mathbb{Q}(E_N \cap \{B_p \leq X_N < B_{ac}\}) \\ & + D(T)\,\mathbb{E}^{\mathbb{Q}}\big[X_N\,\mathbf{1}_{E_N \cap \{X_N < B_p\}}\big]. \end{aligned} \tag{7}$$

The first term corresponds to an early redemption at date $t_i$, the second to a redemption with coupons at maturity, the third to the repayment of the notional only, and the last one to the capital loss scenario.

## 5.2 Linearity in the coupon and embedded put

We now turn to the determination of the fair coupon level $c$. A basic Athena has the convenient property that its price is affine in the coupon. Indeed, the coupon only affects the *amount* paid at redemption, never the redemption time $\tau$, which depends on the path of the underlying alone. Moreover, on the event $E_N$ that the product has not been called before maturity, the terminal payoff can be rewritten, since $B_p < 1$, as

$$\Pi_T = 1 + Nc\,\mathbf{1}_{\{X_N \geq B_{ac}\}} - (1 - X_N)^+\,\mathbf{1}_{\{X_N < B_p\}}. \tag{8}$$

The last term is the payoff of a down-and-in put with strike 100% and barrier $B_p$ observed at maturity, which is sold by the investor. Substituting (8) into (7) yields the decomposition

$$P_0 = \underbrace{\mathbb{E}^{\mathbb{Q}}\big[D(\tau)\big]}_{\text{notional}} + c \underbrace{\sum_{i=1}^{N} i\,D(t_i)\,\mathbb{Q}(E_i \cap \{X_i \geq B_{ac}\})}_{B\ \text{(coupons)}} - \underbrace{D(T)\,\mathbb{E}^{\mathbb{Q}}\Big[(1 - X_N)^+\,\mathbf{1}_{\{X_N < B_p\}}\,\mathbf{1}_{E_N}\Big]}_{\text{PDI}^{ac}}. \tag{9}$$

The investor is thus long a note repaying the notional at the random time $\tau$, long a strip of conditional coupons, and short a down-and-in put $\text{PDI}^{ac}$. Grouping the terms that do not depend on the coupon, the price takes the affine form

$$P_0 = A + c\,B, \qquad A = \mathbb{E}^{\mathbb{Q}}\big[D(\tau)\big] - \text{PDI}^{ac}, \tag{10}$$

where $A$ is the price of the same product with a zero coupon, and $B$ the value of a unit coupon paid in all scenarios where coupons are received.

### 5.3 Fair coupon

The fair coupon $c^\star$ is the coupon for which the price equals a target $P^\star$, namely $P^\star = 1$ for a fair price, or $P^\star = 1 - m$ when the issuer retains a margin $m$. By linearity, it is given in closed form by

$$\boxed{c^\star = \frac{P^\star - A}{B} = \frac{P^\star - \mathbb{E}^{\mathbb{Q}}\big[D(\tau)\big] + \mathrm{PDI}^{ac}}{B}} \tag{11}$$

To determine the root of this equation, a single Monte Carlo simulation provides $A$ and $B$, hence $c^\star$. This expression also makes the economics of the product transparent: the coupons finance the down-and-in put sold by the investor, together with the funding benefit of the issuer, since $\mathbb{E}^{\mathbb{Q}}[D(\tau)] < 1$ as soon as cash flows are discounted with a positive rate and funding spread.

### 5.4 Remarks on the embedded put

The put $\mathrm{PDI}^{ac}$ is not a vanilla barrier option: it is knocked out whenever the product is autocalled, which is captured by the indicator $\mathbf{1}_{E_N}$. Since $\mathbf{1}_{E_N} \leq 1$, its value is bounded by that of the standard European down-and-in put,

$$\mathrm{PDI}^{ac} \leq \mathrm{PDI} = D(T)\,\mathbb{E}^{\mathbb{Q}}\Big[(1 - X_N)^+\,\mathbf{1}_{\{X_N < B_p\}}\Big]. \tag{12}$$

The latter itself decomposes into a vanilla put struck at the barrier and a digital put,

$$(1 - X_N)^+\,\mathbf{1}_{\{X_N < B_p\}} = (B_p - X_N)^+ + (1 - B_p)\,\mathbf{1}_{\{X_N < B_p\}}, \tag{13}$$

so that, absent the autocall feature, its value follows directly from the implied volatility surface around strike $B_p$, the digital term making it sensitive to the skew.

### 5.5 Structuration :

From the investor's perspective, the basic Athena note is composed of:

- a long position in an *autocallable zero-coupon note* with maximal maturity $T$, which repays the notional together with the accrued coupons at the first observation date $t_i$ such that $X_i \geq B_{ac}$, and the notional alone at maturity otherwise;
- a short position in a *down-and-in put* (PDI) with strike 100% and barrier $B_p$ observed at maturity, which is knocked out whenever the product is autocalled.

### 5.6 Sensitivity analysis :

We will now look at the different sensitivities of the product and what are the impacts of a move on certain parameters on $c$. We are now thinking from the investor point of view and with an already contracted note.

#### Volatility

The investor is short the $\mathrm{PDI}^{ac}$, so he is short vega: the price of a down-and-in put increases with the volatility, since a higher volatility makes it more likely that the index ends below the protection barrier. An increase of the volatility therefore decreases the value of the note. If the product is not contracted yet, the put sold by the investor is more valuable and finances a higher coupon $c$. The call probabilities move much less: a higher variance slightly lowers the median of the index, hence the probability to be called at the first dates.

### Protection barrier

If the protection barrier is higher, then it is more probable to cross it and it reduces the price of the note. The effect on the coupon $c$ is that it is increased.

### Recall Barrier

If the recall barrier is increased, it is less probable to be recalled before maturity: the note lives longer and is more often exposed to the put at maturity, so it increases the coupon $c$.

### Dividend

If the underlying asset increases its dividend, every forward $F(0,t)$ decreases: under $\mathbb{Q}$ the index drifts down, the note is called less often and the knock-in is more likely. The investor, who does not receive the dividends of the index, is short dividends: the value of the note decreases and the coupon $c$ increases.

### Spread

The issuer funds itself at its own rate $r+s$, where $s$ is its credit spread, and discounts the cash flows of the note at this rate. A higher spread lowers $\mathbb{E}[D(\tau)]$: the funding benefit $1-\mathbb{E}[D(\tau)]$ of the issuer, which appears in the numerator of (11), increases and is given back to the investor through a higher coupon $c$. The counterpart is the credit risk of the issuer, borne by the investor.

### Risk free rate

A higher risk-free rate has two opposite effects. It raises the forwards, so the note is called earlier and the knock-in is less likely, which lowers the value of the put and the coupon. But it also lowers the discount factors: the funding benefit $1-\mathbb{E}[D(\tau)]$ increases, which raises the coupon. For the base case the second effect dominates, and the coupon $c$ increases with the rates.

### Maturity

A longer maximal maturity leaves more time to the paths that are not called to end below the protection barrier: the put is more valuable and the coupon $c$ increases. The effect decreases with the maturity, since most notes are called during the first year and the additional premium is spread over a longer expected life.

### Skew

The put decomposes into a vanilla put struck at the barrier and a digital put; the digital part is the derivative of the vanilla price with respect to the strike, so it is directly priced by the skew. A steeper (more negative) skew makes the low strikes more expensive and should raise the coupon $c$. But the skew also changes the distribution of the index: with a negative skew, the distribution is more left-skewed and its median is higher, so the note is called more often at the first dates and lives shorter. For an Athena the two effects largely offset each other.

### Correlation

On a single underlying, the only correlation is the one between the index and its volatility, which drives the skew. For a worst-of Athena on several underlyings, the investor is long correlation: a lower correlation makes it more likely that one of the underlyings ends below the protection barrier, and less likely that all of them are above the autocall barrier at the same time. The put is more valuable, the call less likely, and the coupon $c$ increases. This is why worst-of notes pay higher coupons; they are outside the scope of this report (Section 7).

## 5.7 Pricing : A numerical example

Let us now consider a synthetic Athena written on the CAC 40 index (dividends not reinvested). Its initial level is fixed on 12 February 2025, the date of the market data, and its maximal maturity is 4 years. A coupon $c$ of 6% per annum accrues every quarter and is paid upon redemption (snowball). The underlying is observed quarterly: at each observation date, the product is redeemed early if the index closes at or above 100% of its initial level. If the product has not been called, the investor receives at maturity the notional plus the accrued coupons if the index is at or above its initial level, the notional alone if it is at or above the protection barrier $B_p = 60\%$, and the notional reduced by the full decline of the index otherwise. The base case is priced without the credit spread of the issuer, whose impact is studied with the other market parameters below.

### 5.7.1 Model

The underlying follows the Heston model of Section 3, calibrated on the CAC 40 implied volatility surface of 12 February 2025, with the forward curve bootstrapped from put–call parity. The paths are simulated on a weekly grid with 50,000 paths (Section 4); a daily grid is used for the daily observations and the daily monitored protection. All the figures and tables of this section come from the notebook `Products/Athena/athena_synthetic_pricing.ipynb`.

### 5.7.2 Pricing

For the purpose of the pricing of this product, we will use the script `athena.py`[1] to build the product and `monte_carlo.py`[2] to simulate the different paths.

In this first basic case, we have the following result:

We also plotted the distribution of call probability over the different paths.

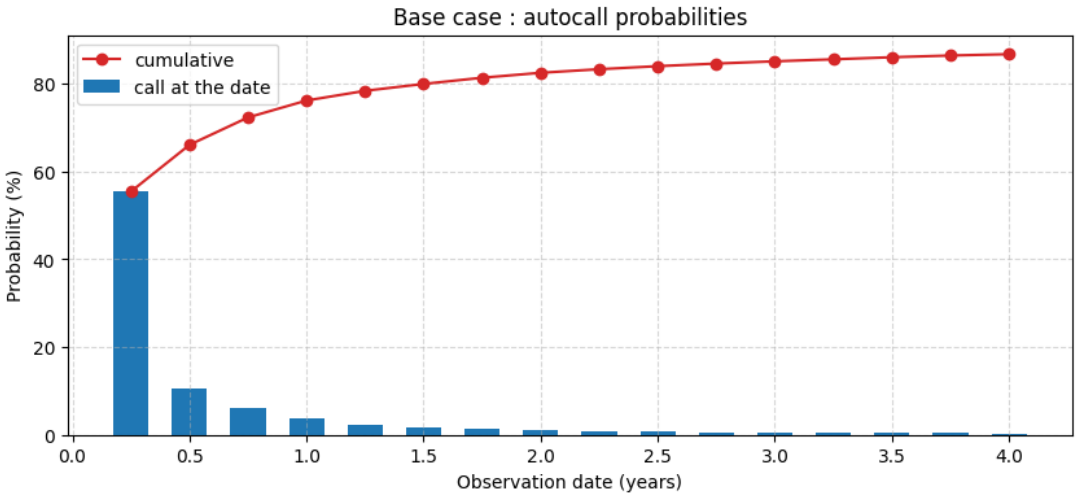



Figure 2: Distribution of the call probability

As we can see, the probability for the product to be called during the first observation period is around 55% and the cumulative probability of call is around 86%.

We then present the distribution of the different paths:

The expected life of this product is pretty short (1.03 years for a maximal maturity of 4 years, the median life being three months), and the coupon actually received is only 3.01% of the nominal on average, far below $4 \times 6\%$. Because the investor is short a PDI, the capital loss is

[1] https://github.com/benguedalauzonelias/portfolio/tree/main/Products/Athena
[2] https://github.com/benguedalauzonelias/portfolio/tree/main/Equity_Models

| | | **Base case** |
|---|---|---|
| Contract | Maturity (years) | 4 |
| | Number of observations | 16 |
| | Observation frequency (years) | 0.25 |
| | Coupon (% p.a.) | 6 |
| | Coupon type | Snowball |
| | First autocall barrier (%) | 100 |
| | Last autocall barrier (%) | 100 |
| | Step-down per observation (%) | 0 |
| | First call date (years) | 0.25 |
| | Protection barrier (%) | 60 |
| | Protection type | European |
| | Redemption if knocked-in | Standard |
| | Capital guaranteed (%) | 0 |
| | Buffer (%) | 0 |
| | Range accrual (%) | – |
| | Duo barrier (%) | – |
| | Duo coupon ratio | – |
| | Look-back | – |
| | Nominal | 100 |
| Pricing | Price (% nominal) | 97.80 |
| | 95% confidence interval (±) | 0.11 |
| | Fair coupon (% p.a.) | 10.54 |
| Scenarios | Probability to reach maturity (%) | 13.29 |
| | Probability of a capital loss (%) | 5.22 |
| | Expected loss at maturity (% nominal) | 3.05 |
| | Expected life (years) | 1.03 |
| | Expected number of coupons | 0.87 |
| | Expected coupon paid (% nominal) | 3.01 |

Table 4: Contract features, pricing results and scenario statistics of the base-case Athena.

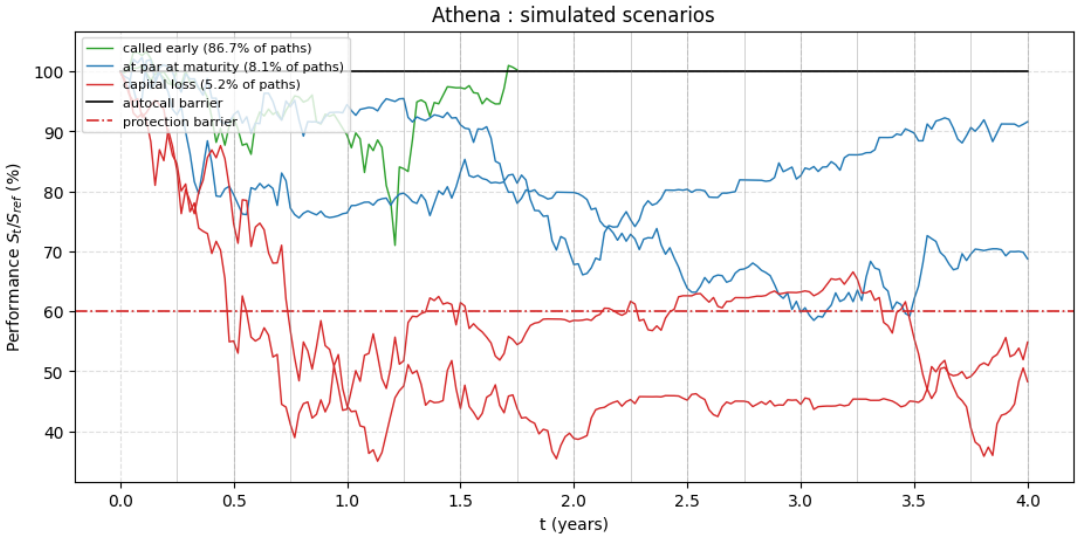



Figure 3: Distribution of the different paths

rare (5.22%) but large when it happens: the redemption jumps from 100 to 60 at the protection barrier, and the expected loss given a loss is about 58% of the nominal.

### 5.7.3 Sensitivities

We are now interested in how the different parameters impact the price of a basic Athena product. We notice that the price after the emission of the note and the fair-coupon $c^\star$ move in opposite directions: a change that lowers the value of the note for a given coupon must be paid back by a higher coupon at issuance. Each parameter is moved alone, all the others being those of the base case, and the paths are the same for every scenario (common random numbers), so that the differences are not blurred by the Monte Carlo noise.

### Protection barrier

We plotted the different parameters against a move of the protection barrier:

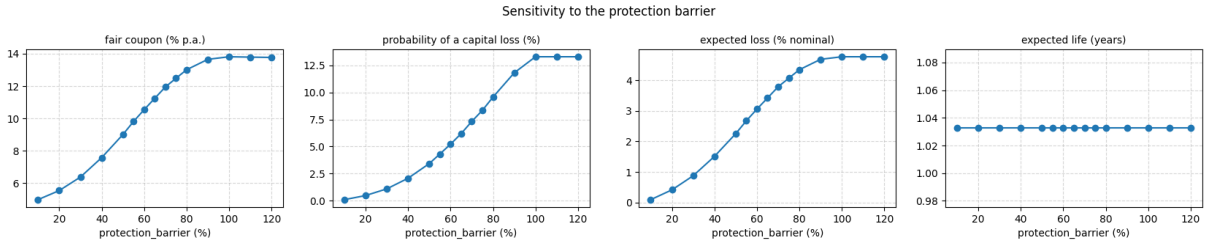



Figure 4: Fair coupon, probability of a capital loss, expected loss and expected life as functions of the protection barrier.

As we can see, an increase of the protection barrier increases the value of $c^\star$. It is easy to understand: a higher protection barrier exposes the investor more often to a capital loss at maturity, so the down-and-in put that he sells is more valuable, and he must be remunerated for this risk. The probability of a capital loss and the expected loss increase in the same direction as the barrier. The call probabilities, hence the expected life, do not move: the protection only matters at maturity, for the paths that were never called. Above 100%, nothing changes anymore: since the put is struck at 100%, a knocked-in path that ends above its initial level is redeemed at par anyway.

| Protection barrier (%) | Fair coupon (% p.a.) | Probability of a capital loss (%) | Expected loss (% nominal) |
|---|---|---|---|
| 10 | 4.99 | 0.10 | 0.09 |
| 20 | 5.55 | 0.48 | 0.41 |
| 30 | 6.38 | 1.09 | 0.87 |
| 40 | 7.58 | 2.07 | 1.50 |
| 50 | 9.00 | 3.43 | 2.25 |
| 55 | 9.81 | 4.32 | 2.67 |
| **60** | **10.54** | **5.22** | **3.05** |
| 65 | 11.24 | 6.19 | 3.42 |
| 70 | 11.94 | 7.32 | 3.78 |
| 75 | 12.48 | 8.34 | 4.07 |
| 80 | 13.00 | 9.57 | 4.34 |
| 90 | 13.64 | 11.81 | 4.68 |
| 100 | 13.80 | 13.29 | 4.77 |

Table 5: Fair coupon and loss statistics as a function of the protection barrier, all other parameters being those of the base case (in bold). The expected life of the product, equal to 1.03 years, does not depend on the protection barrier.

### Autocall barrier

We also plotted the reaction of the different parameters for a move of the autocall barrier.

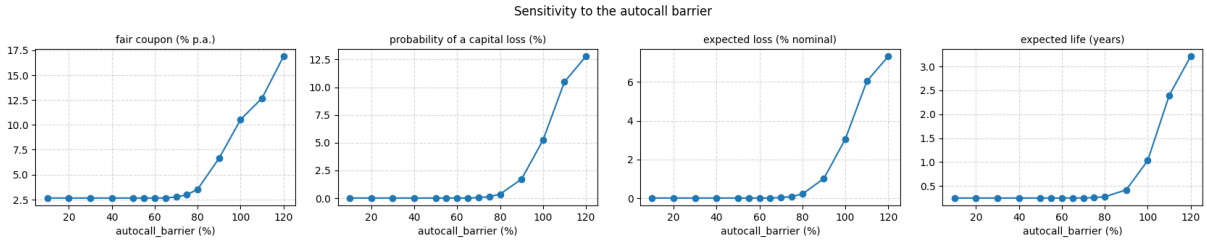



Figure 5: Fair coupon, probability of a capital loss, expected loss and expected life as functions of the autocall barrier.

| Autocall barrier (%) | Fair coupon (% p.a.) | Probability of a capital loss (%) | Expected loss (% nominal) | Expected life (years) |
|---|---|---|---|---|
| 50 | 2.65 | 0.00 | 0.00 | 0.25 |
| 60 | 2.65 | 0.00 | 0.00 | 0.25 |
| 70 | 2.75 | 0.04 | 0.02 | 0.25 |
| 80 | 3.54 | 0.35 | 0.21 | 0.28 |
| 90 | 6.66 | 1.70 | 1.01 | 0.42 |
| **100** | **10.54** | **5.22** | **3.05** | **1.03** |
| 110 | 12.69 | 10.48 | 6.04 | 2.39 |
| 120 | 16.89 | 12.79 | 7.33 | 3.21 |

Table 6: Fair coupon, loss statistics and expected life as a function of the autocall barrier, all other parameters being those of the base case (in bold).

A higher autocall barrier makes the call less likely: the note lives longer (from 0.42 years at 90% to 2.39 years at 110%), more paths reach maturity and are exposed to the put, and the fair coupon increases steeply, from 6.66% at 90% to 12.69% at 110%. The low barriers give a useful limit case: below 60%, the note is called at the first date on every path, so that it is nothing but a three-month deposit. The fair coupon is then $c^\star = (1/D(t_1) - 1)/t_1 \approx 2.65\%$, the three-month rate: without put and without any risk of a long life, the investor only receives the funding rate.

### Market parameters

The note is now kept unchanged (coupon 6%, barriers of the base case) and the market moves. For each scenario we report the price of the note at a coupon of 6%, the fair coupon, the probability of a call at the first date, the probability of a capital loss and the expected life.

**Volatility.** $\sqrt{v_0}$ and $\sqrt{\theta}$ are shifted together by $-3$ to $+3$ volatility points (Figure 6, Table 7). One volatility point costs about 0.33 on a nominal of 100 and adds about 0.65 point of fair coupon: the investor is short the down-and-in put, whose value increases with the volatility (the probability of a capital loss goes from 3.79% to 6.70%). The call probabilities move little: a higher variance lowers the median of $S_t$, so the probability to be above 100% at the first date decreases slightly (from 56.8% to 54.2%) and the expected life increases slightly.

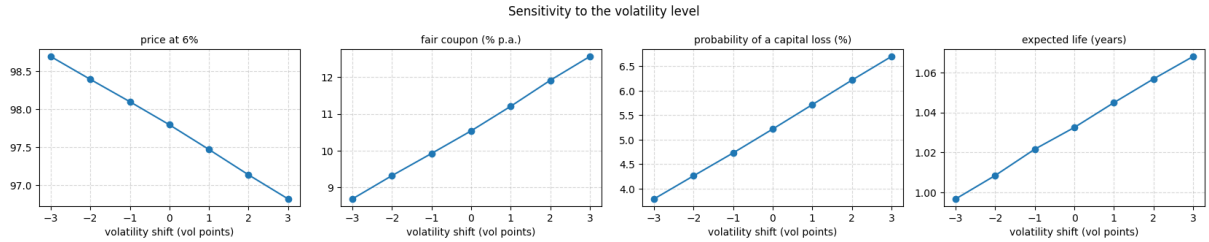



Figure 6: Base case: price at 6%, fair coupon, probability of a capital loss and expected life as functions of a parallel shift of the volatility level.

**Dividends.** A higher borrow + dividend yield lowers every forward $F(0,t)$: the index drifts down under $\mathbb{Q}$, the note is called less often (expected life from 0.96 to 1.11 years for a shift from $-1\%$ to $+1\%$) and the knock-in is more likely (4.51% to 6.07%). The investor, who does not receive the dividends of the index, is short dividends: one point of dividend yield costs about 0.6 on 100 and adds about 1.3 points of coupon, twice the effect of one volatility point.

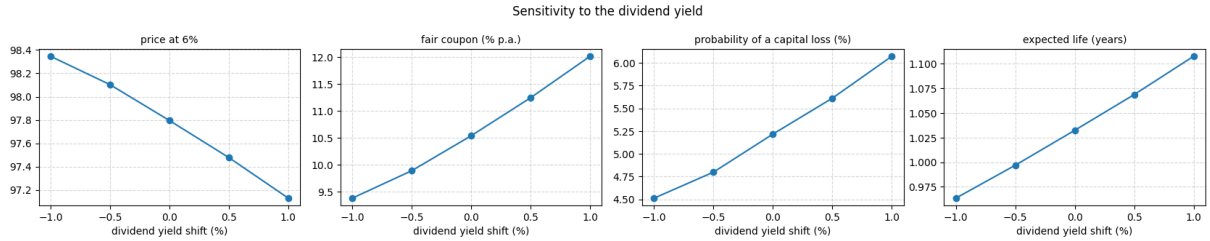



Figure 7: Base case as a function of a parallel shift of the borrow + dividend yield.

**Risk-free rate.** A parallel shift of the zero-coupon curve has two opposite effects. The forwards move by $e^{\Delta r\, T}$: the note is called earlier and the knock-in is less likely, which lowers the value of the put. But the cash flows are also discounted more, and the funding benefit of the issuer $1 - \mathbb{E}[D(\tau)]$ increases. For the base case the second effect dominates: the price at 6% decreases with the rates (98.09 at $-1\%$, 97.51 at $+1\%$) and the fair coupon increases from 9.95% to 11.16%. This is why the coupons of autocallable notes rise with the level of interest rates.

**Issuer spread.** The base case is priced without credit risk. A real issuer funds itself at $r + s$, $s$ being its credit spread, and discounts the cash flows of the note at this rate, while the drift of the index is unchanged. The scenarios do not move, but every cash flow is worth less: each 50bp of spread costs about 0.44 on 100 and adds about 0.96 point of fair coupon (from 10.57%

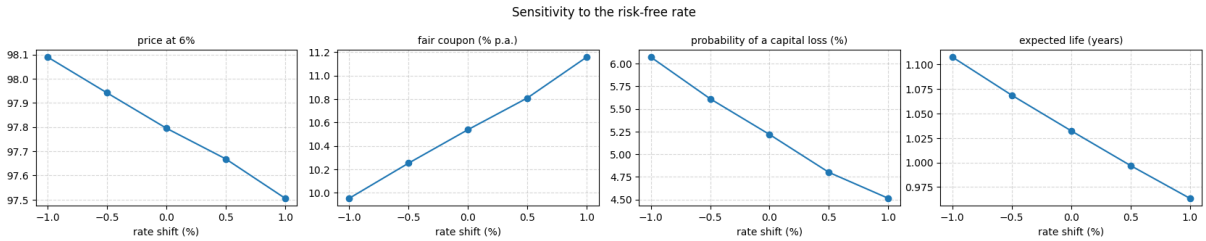



Figure 8: Base case as a function of a parallel shift of the zero-coupon curve.

at $s = 0$ to 14.40% at $s = 200$bp, the control variate being switched off for this sweep). A riskier issuer can therefore offer a higher coupon, the counterpart being the credit risk borne by the investor.

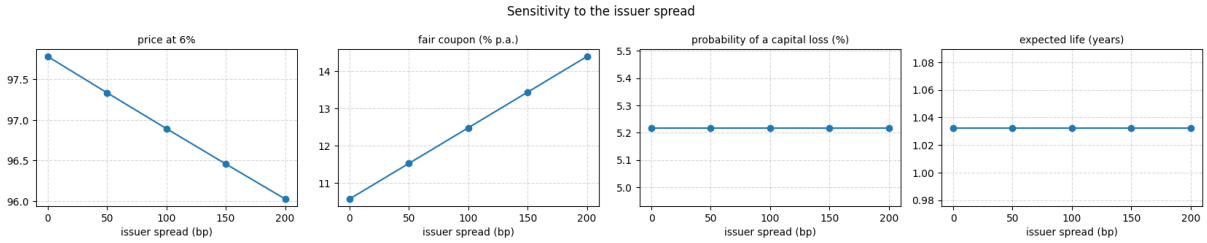



Figure 9: Base case as a function of the credit spread of the issuer.

**Maturity.** With a longer maximal maturity, the paths that are not called early have more time to fall below the protection barrier (probability of a loss from 1.67% at 1 year to 5.62% at 5 years), and the put is more valuable. The fair coupon increases with the maturity, but at a decreasing rate (7.56%, 9.50%, 10.21%, 10.54% and 10.60% from 1 to 5 years): most notes are called in the first year whatever the maturity, and the additional put premium is spread over a longer expected life.

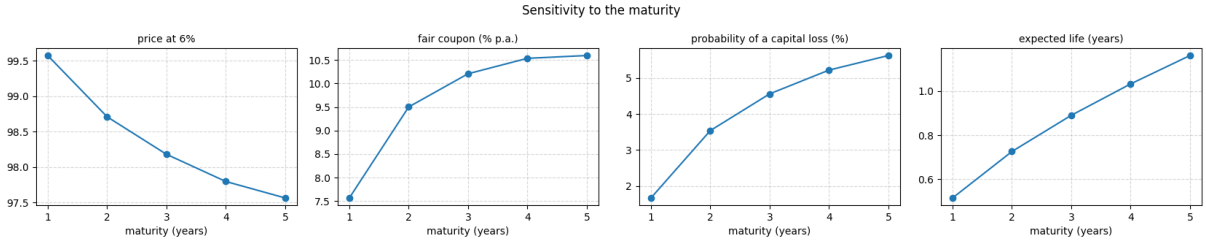



Figure 10: Same note with a maximal maturity from 1 to 5 years.

**Skew.** In the Heston model, the skew of the implied volatility surface is mainly driven by the spot-variance correlation $\rho$, calibrated at $-0.605$. We move $\rho$ alone, from $-0.9$ to 0, the variance level being unchanged. A more negative $\rho$ steepens the skew: the 4-year European put struck at 60% is worth 2.39% of $S_0$ at $\rho = -0.9$ against 1.54% at $\rho = 0$, and the digital part of the put sold by the investor is more expensive. But a more negative skew also makes the distribution of $S_t$ more left-skewed, with a higher median: the note is more often called at the first date (60.0% at $\rho = -0.9$ against 46.9% at $\rho = 0$) and its expected life is shorter. For the Athena, the two effects offset each other: the fair coupon stays between 10.3% and 10.6% over the whole range. The skew is a risk on the distribution of the scenarios more than on the price.

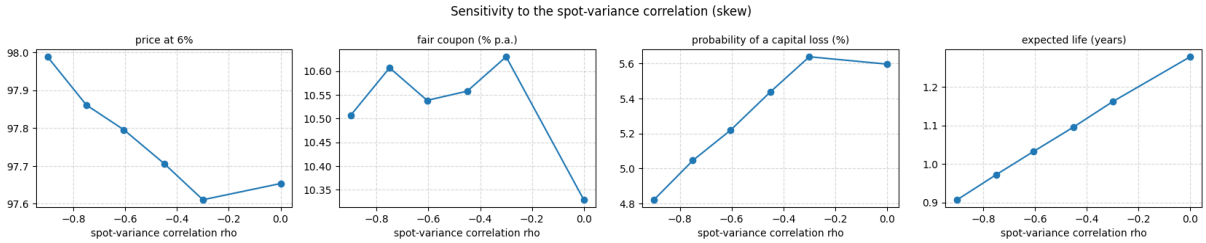



Figure 11: Base case as a function of the spot-variance correlation $\rho$ of the Heston model.

| | **Scenario** | **Price at 6%** | **Fair coupon (% p.a.)** | **P(call $t_1$) (%)** | **P(loss) (%)** | **E[life] (y)** |
|---|---|---|---|---|---|---|
| Volatility | $-3$ vol pts | 98.70 | 8.69 | 56.82 | 3.79 | 1.00 |
| | base | 97.80 | 10.54 | 55.43 | 5.22 | 1.03 |
| | $+3$ vol pts | 96.82 | 12.56 | 54.17 | 6.70 | 1.07 |
| Dividend yield | $-1\%$ | 98.35 | 9.38 | 56.95 | 4.51 | 0.96 |
| | $+1\%$ | 97.13 | 12.02 | 53.88 | 6.07 | 1.11 |
| Rates | $-1\%$ | 98.09 | 9.95 | 53.88 | 6.07 | 1.11 |
| | $+1\%$ | 97.51 | 11.16 | 56.95 | 4.51 | 0.96 |
| Issuer spread | 0bp | 97.78 | 10.57 | 55.43 | 5.22 | 1.03 |
| | 100bp | 96.89 | 12.48 | 55.43 | 5.22 | 1.03 |
| | 200bp | 96.03 | 14.40 | 55.43 | 5.22 | 1.03 |
| Maturity | 1 year | 99.57 | 7.56 | 55.43 | 1.67 | 0.52 |
| | 2 years | 98.71 | 9.50 | 55.43 | 3.54 | 0.73 |
| | 5 years | 97.56 | 10.60 | 55.43 | 5.62 | 1.16 |
| Correlation $\rho$ | $-0.9$ | 97.99 | 10.51 | 59.97 | 4.82 | 0.91 |
| | 0 | 97.65 | 10.33 | 46.89 | 5.60 | 1.28 |

Table 7: Base case under moved market parameters (coupon 6%, nominal 100). Base case: price 97.80, fair coupon 10.54%.

### 5.7.4 Hedging

If we think from the desk's point of view, the issuer is short the note: it is long the down-and-in put sold by the investor and short the autocall feature. Its main sensitivity to manage is the delta, which it hedges with futures on the index; it must also manage the vega, which it hedges with listed puts, and the dividends, which it hedges with dividend futures. Table 8 gives the hedge of the base case at inception, for a nominal of 1 million euros (10,000 notes of 100).

| **Risk of the note** (nominal 100) | **Value** | **Hedge of the issuer** (nominal 1M) |
|---|---|---|
| Delta | 0.00470 per index point | buy 47 index units ($\approx$ 378,000 EUR of futures) |
| Vega | $-0.315$ per vol point | long 3,150 EUR per vol point: sell 163 puts 4 years, strike 60% |
| Dividend | $-0.689$ per $+1\%$ | long 6,890 EUR per point of yield: sell dividend futures |
| Rho | $-0.210$ per $+1\%$ | long 2,100 EUR per point of rate: receive fixed on a swap |

Table 8: Base case: Greeks of the note and hedge of the issuer at inception. The vega of the 4-year put struck at 60% is 19.3 per vol point and per unit of index.

The note is equivalent to 37.8% of its nominal invested in the index: the delta is far below one because the investor only bears the downside below the protection barrier and gives up the upside above the autocall barrier. This exposure is not stable (Figure 12, the initial fixing staying at $S_0 = 8042$): when the index falls towards the protection barrier, the call becomes unlikely and the put gets closer to the money, so the delta of the note increases from 0.0046 at $S_0$ to 0.0146 at the barrier (4825), and the issuer must buy index in a falling market. The vega of the note is most negative between 60% and 70% of the initial level, where the knock-in is most uncertain. Close to maturity, the jump of the redemption at the protection barrier (the digital part of the put) makes the delta and the gamma very large around the barrier: this pin risk is the main difficulty of the hedge, and the reason why issuers often prefer an airbag or a buffer, whose payoff is continuous at the barrier.

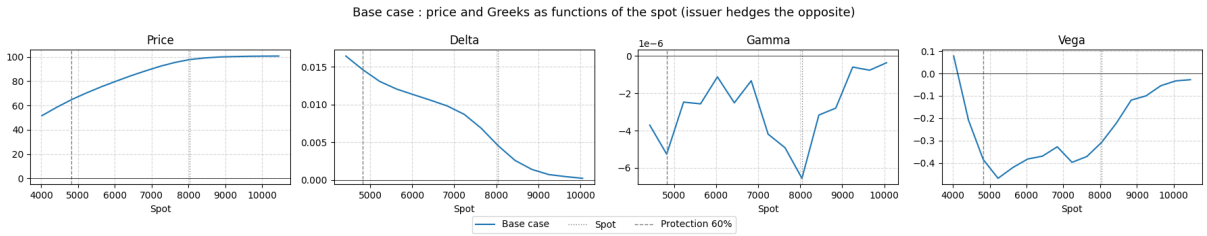



Figure 12: Base case: price, delta, gamma and vega of the note as functions of the spot (30,000 paths). The issuer holds the opposite of these Greeks.

For real hedging of this kind of product, have a look to the paper pricing of barrier options[3].

[3] https://portfolio-eliasbenguedda.fr/projects/barriere_option.html

### 5.8 Features of Athena products

We have seen the basic Athena product, its construction, its sensitivities and its hedging. Now, the investor could want to have better guarantees or a higher coupon. For this purpose, we can add some interesting features to the product that we will expose in the following parts of this paper. Each feature is one parameter of the class `Athena` (Table 1), all the other terms being those of the base case.

### 5.8.1 Exercise frequency

First thing first, the exercise frequency can be changed. Instead of looking at the level of the underlying every quarter, one could want to observe it every year, every month, or even every day. Table 9 gives the results from annual to daily observations.

| | | **Base case** | **Annual** | **Semi-annual** | **Monthly** | **Daily** |
|---|---|---|---|---|---|---|
| Contract | Number of observations | 16 | 4 | 8 | 48 | 1008 |
| | Observation frequency (years) | 0.25 | 1 | 0.5 | 1/12 | 1/252 |
| Pricing | Price (% nominal) | 97.80 | 96.61 | 97.22 | 98.62 | 99.65 |
| | 95% confidence interval (±) | 0.11 | 0.12 | 0.12 | 0.09 | 0.05 |
| | Fair coupon (% p.a.) | 10.54 | 9.41 | 9.93 | 11.11 | 11.74 |
| Scenarios | Probability to reach maturity (%) | 13.29 | 25.70 | 18.61 | 7.77 | 1.80 |
| | Probability of a capital loss (%) | 5.22 | 9.06 | 6.97 | 3.13 | 0.78 |
| | Expected loss at maturity (% nominal) | 3.05 | 5.27 | 4.08 | 1.83 | 0.46 |
| | Expected life (years) | 1.03 | 2.06 | 1.48 | 0.59 | 0.13 |
| | Expected number of coupons | 0.87 | 0.74 | 0.81 | 0.92 | 0.98 |
| | Expected coupon paid (% nominal) | 3.01 | 6.22 | 4.39 | 1.66 | 0.38 |

Table 9: Effect of the observation frequency of the autocall barrier, all other parameters being those of the base case.

More frequent observations call the note earlier: the expected life falls from 2.06 years (annual) to 0.13 year (daily), and the probability of a capital loss from 9.06% to 0.78%. One could expect a lower coupon, since the investor takes less risk. It is the opposite: the fair coupon increases from 9.41% to 11.74%. The snowball coupon $c\,t_k$ is paid on the life of the note, and with daily observations the note is usually called within a few weeks: the investor receives only 0.38% of coupon on average, against 6.22% with annual observations. The put premium, smaller in value, is spread over a much shorter life, which requires a higher coupon rate per annum.

### 5.8.2 Step-down barrier

The step down feature is really easy to understand. In fact, periodically, the autocall barrier is moving. It can be in both directions, but generally the barrier is decreasing over time. We took the example of a barrier decreasing by 1% at each observation date: at the last observation date, the barrier is at 85% of the initial level. We have the following result:

| | | **Base case** | **Step-down 1%** |
|---|---|---|---|
| Contract | Last autocall barrier (%) | 100 | 85 |
| | Step-down per observation (%) | 0 | 1 |
| Pricing | Price (% nominal) | 97.80 | 98.81 |
| | 95% confidence interval (±) | 0.11 | 0.10 |
| | Fair coupon (% p.a.) | 10.54 | 8.24 |
| Scenarios | Probability to reach maturity (%) | 13.29 | 8.12 |
| | Probability of a capital loss (%) | 5.22 | 4.44 |
| | Expected loss at maturity (% nominal) | 3.05 | 2.62 |
| | Expected life (years) | 1.03 | 0.87 |
| | Expected number of coupons | 0.87 | 0.92 |
| | Expected coupon paid (% nominal) | 3.01 | 3.29 |

Table 10: Effect of a step-down autocall barrier, decreasing by 1% per quarterly observation from 100% to 85%, all other parameters being those of the base case.

The lower barriers of the last dates call the notes that were drifting slightly below their initial level: the probability to reach maturity falls from 13.29% to 8.12%, and these paths, which were the closest to the protection barrier, are no longer exposed to the put. The note is worth more to the investor (98.81 instead of 97.80 at 6%) and the fair coupon falls to 8.24%. The step-down is a way to sell a note with a higher probability of success, at the cost of a lower coupon. Note that a call at a barrier below 100% pays the coupon even if the index has lost up to 15%.

### 5.8.3 Airbag

The airbag effect is also easy to understand: the put sold by the investor is struck at the protection barrier instead of 100%. Below 60%, the investor receives $NP_T/B^{\text{prot}}$ instead of $NP_T$: the redemption is continuous at the barrier, and an index at 50% is redeemed at 83.3 instead of 50 (Figure 1).

| | | **Base case** | **Airbag** |
|---|---|---|---|
| Contract | Put strike (%) | 100 | 60 |
| Pricing | Price (% nominal) | 97.80 | 99.11 |
| | 95% confidence interval (±) | 0.11 | 0.07 |
| | Fair coupon (% p.a.) | 10.54 | 7.83 |
| Scenarios | Probability to reach maturity (%) | 13.29 | 13.29 |
| | Probability of a capital loss (%) | 5.22 | 5.22 |
| | Expected loss at maturity (% nominal) | 3.05 | 1.61 |
| | Expected life (years) | 1.03 | 1.03 |
| | Expected number of coupons | 0.87 | 0.87 |
| | Expected coupon paid (% nominal) | 3.01 | 3.01 |

Table 11: Airbag: the put is struck at the protection barrier (60%), all other parameters being those of the base case.

The scenarios are exactly those of the base case: only the amount lost changes. The expected loss is halved (1.61% of the nominal instead of 3.05%), the probability of a loss being unchanged, and the fair coupon falls from 10.54% to 7.83%. The short put is now a put struck at 60% on $N/B^{\text{prot}}$ units, without the digital part $(1 - B^{\text{prot}})\,\mathbf{1}_{\{P_T < B^{\text{prot}}\}}$: the airbag removes the jump of the payoff, hence most of the skew and pin risks of the issuer.

### 5.8.4 Buffer

With a buffer $b$, the first $b$ of decline of the index is absorbed by the issuer, and the investor loses one for one below $1-b$: the redemption is $N\min(1, P_T + b)$ below the protection barrier $B^{\text{prot}} = 1 - b$. We take a buffer of 10%.

| | | **Base case** | **Buffer 10%** |
|---|---|---|---|
| Contract | Protection barrier (%) | 60 | 90 |
| | Buffer (%) | 0 | 10 |
| Pricing | Price (% nominal) | 97.80 | 97.37 |
| | 95% confidence interval (±) | 0.11 | 0.10 |
| | Fair coupon (% p.a.) | 10.54 | 11.42 |
| Scenarios | Probability to reach maturity (%) | 13.29 | 13.29 |
| | Probability of a capital loss (%) | 5.22 | 11.81 |
| | Expected loss at maturity (% nominal) | 3.05 | 3.50 |
| | Expected life (years) | 1.03 | 1.03 |
| | Expected number of coupons | 0.87 | 0.87 |
| | Expected coupon paid (% nominal) | 3.01 | 3.01 |

Table 12: Buffer of 10%: the investor only loses below 90%, one for one, all other parameters being those of the base case.

A buffer of 10% protects the investor less than a barrier at 60%: the losses start as soon as the index ends below 90%, so the probability of a capital loss rises from 5.22% to 11.81% and the fair coupon from 10.54% to 11.42%. But there is no more loss jump: an index at 55% is redeemed at 65 instead of 55, and the largest losses are reduced by 10 points. The buffer trades a higher probability of small losses for smaller large losses; the issuer, short a put spread instead of a put with a digital, has a much smoother hedge.

### 5.8.5 Guaranteed capital

The capital is guaranteed by flooring the redemption at maturity at $\phi N$: with $\phi = 100\%$, the investor never loses capital, with $\phi = 90\%$ he loses at most 10%.

| | | **Base case** | **Guaranteed 90%** | **Guaranteed 100%** |
|---|---|---|---|---|
| Contract | Capital guaranteed (%) | 0 | 90 | 100 |
| Pricing | Price (% nominal) | 97.80 | 100.09 | 100.57 |
| | 95% confidence interval (±) | 0.11 | 0.05 | 0.04 |
| | Fair coupon (% p.a.) | 10.54 | 5.81 | 4.83 |
| Scenarios | Probability to reach maturity (%) | 13.29 | 13.29 | 13.29 |
| | Probability of a capital loss (%) | 5.22 | 5.22 | 0.00 |
| | Expected loss at maturity (% nominal) | 3.05 | 0.52 | 0.00 |
| | Expected life (years) | 1.03 | 1.03 | 1.03 |
| | Expected number of coupons | 0.87 | 0.87 | 0.87 |
| | Expected coupon paid (% nominal) | 3.01 | 3.01 | 3.01 |

Table 13: Capital guarantee: the redemption at maturity is floored at 90% or 100% of the nominal.

With a full guarantee, the put disappears: the note is a zero-coupon bond plus the autocall feature, financed only by the funding benefit $1 - \mathbb{E}[D(\tau)]$, and the fair coupon falls to 4.83%. With 90% guaranteed, the investor is only short the put spread between 100% and 90%: the probability of a loss is unchanged (5.22%), but the expected loss is divided by six (0.52% instead of 3.05%), for a fair coupon of 5.81%. In the decomposition $P_0 = A + cB$, the guarantee removes all or most of $\text{PDI}^{ac}$ from $A$, which leaves little to finance the coupon. These notes are mostly sold when rates are high, since the funding benefit is then larger.

### 5.8.6 Coupon range accrual

With a range accrual coupon, the snowball coupon only accrues on the days where the index is inside a range: a call at $t_k$ pays $c$ times the time spent in the range since inception, instead of $c\,t_k$. We take a range of $[80\%, 120\%]$, observed on the weekly simulation dates.

| | | **Base case** | **Range accrual 80-120%** |
|---|---|---|---|
| Contract | Range accrual (%) | – | [80, 120] |
| Pricing | Price (% nominal) | 97.80 | 97.68 |
| | 95% confidence interval (±) | 0.11 | 0.11 |
| | Fair coupon (% p.a.) | 10.54 | 10.99 |
| Scenarios | Probability to reach maturity (%) | 13.29 | 13.29 |
| | Probability of a capital loss (%) | 5.22 | 5.22 |
| | Expected loss at maturity (% nominal) | 3.05 | 3.05 |
| | Expected life (years) | 1.03 | 1.03 |
| | Expected number of coupons | 0.87 | 0.87 |
| | Expected coupon paid (% nominal) | 3.01 | 2.88 |

Table 14: Range accrual: the coupon only accrues on the weekly dates where the index is between 80% and 120%.

The call dates are those of the base case. The coupon paid at the call is reduced on the paths that went above 120% or below 80% before being called (2.88% of coupon on average instead of 3.01%), so the fair coupon increases slightly, from 10.54% to 10.99%. The effect is small here because most notes are called at the first dates, before the index has time to leave the range; it would be larger with a narrow range or a long non-call period.

### 5.8.7 Duo mechanism

The duo mechanism adds a second, lower call level: above 100%, the note is called with the full coupon; between 90% and 100%, it is also called, but with half of the coupon.

| | | **Base case** | **Duo 90% / 50%** |
|---|---|---|---|
| Contract | Duo barrier (%) | – | 90 |
| | Duo coupon ratio | – | 0.5 |
| Pricing | Price (% nominal) | 97.80 | 99.37 |
| | 95% confidence interval (±) | 0.11 | 0.07 |
| | Fair coupon (% p.a.) | 10.54 | 8.86 |
| Scenarios | Probability to reach maturity (%) | 13.29 | 3.13 |
| | Probability of a capital loss (%) | 5.22 | 1.70 |
| | Expected loss at maturity (% nominal) | 3.05 | 1.01 |
| | Expected life (years) | 1.03 | 0.42 |
| | Expected number of coupons | 0.87 | 0.97 |
| | Expected coupon paid (% nominal) | 3.01 | 1.33 |

Table 15: Duo mechanism: call with the full coupon above 100%, with half of the coupon between 90% and 100%.

The note is now called on almost every path (the probability to reach maturity falls from 13.29% to 3.13%), very early (expected life 0.42 year): the investor gives up half of the coupon on the paths called between 90% and 100%, but these paths are no longer exposed to the put. The probability of a capital loss falls from 5.22% to 1.70% and the fair coupon to 8.86%. The duo is a compromise between a step-down and a lower autocall barrier: with a coupon ratio of 100%, it would be exactly an autocall barrier at 90% (fair coupon 6.66%, Table 6).

### 5.8.8 Look back

With a look back, the reference level $S_{\text{ref}}$ is not the initial fixing but the minimum of the index over the first month (a lookback on the average gives an Asian fixing). Every barrier is then a fraction of a lower reference level on the paths where the index fell during the first month.

| | | **Base case** | **Look back min 1m** |
|---|---|---|---|
| Contract | Look back | – | min over 1 month |
| Pricing | Price (% nominal) | 97.80 | 98.75 |
| | 95% confidence interval (±) | 0.11 | 0.10 |
| | Fair coupon (% p.a.) | 10.54 | 9.08 |
| Scenarios | Probability to reach maturity (%) | 13.29 | 9.03 |
| | Probability of a capital loss (%) | 5.22 | 3.57 |
| | Expected loss at maturity (% nominal) | 3.05 | 2.09 |
| | Expected life (years) | 1.03 | 0.78 |
| | Expected number of coupons | 0.87 | 0.91 |
| | Expected coupon paid (% nominal) | 3.01 | 2.51 |

Table 16: Look back: the reference level is the minimum of the index over the first month.

The investor gets the best entry point of the first month: the barriers are lower on the paths that fell, so the note is called more often (probability to reach maturity 9.03% instead of 13.29%) and the capital is lost less often (3.57% instead of 5.22%). The note is worth more to the investor and the fair coupon falls to 9.08%. The look back is the cheapest way to protect an investor against a bad timing of the issue.

### 5.8.9 Protection type

The protection can also be American: the capital is at risk as soon as the index touches 60% at any time before maturity, and not only at maturity. The barrier is monitored continuously (Brownian bridge on the weekly grid) or on daily closes (daily grid).

| | | **Base case** | **American** | **American (daily)** |
|---|---|---|---|---|
| Contract | Protection type | European | Continuous | Daily closes |
| Pricing | Price (% nominal) | 97.80 | 96.83 | 96.92 |
| | 95% confidence interval (±) | 0.11 | 0.12 | 0.12 |
| | Fair coupon (% p.a.) | 10.54 | 12.53 | 12.31 |
| Scenarios | Probability to reach maturity (%) | 13.29 | 13.29 | 13.23 |
| | Probability of a capital loss (%) | 5.22 | 9.22 | 8.98 |
| | Expected loss at maturity (% nominal) | 3.05 | 4.11 | 4.06 |
| | Expected life (years) | 1.03 | 1.03 | 1.03 |
| | Expected number of coupons | 0.87 | 0.87 | 0.87 |
| | Expected coupon paid (% nominal) | 3.01 | 3.01 | 3.01 |

Table 17: American protection at 60%, observed continuously (Brownian bridge) or on daily closes.

A note whose index touched 60% and then recovered is still knocked-in: the probability of a capital loss rises from 5.22% to 9.22% (continuous monitoring), and the fair coupon from 10.54% to 12.53%. The American protection is the most expensive feature for the investor: two points of coupon. A daily monitoring misses the crossings between two closes and is slightly cheaper for the investor (fair coupon 12.31%).

### 5.8.10 Non-call period

During a non-call period, the note cannot be called. With a non-call period of one year, the first callable date is $t = 1$ instead of $t = 0.25$.

| | | **Base case** | **Non-call 1 year** |
|---|---|---|---|
| Contract | First call date (years) | 0.25 | 1 |
| Pricing | Price (% nominal) | 97.80 | 97.15 |
| | 95% confidence interval (±) | 0.11 | 0.12 |
| | Fair coupon (% p.a.) | 10.54 | 8.94 |
| Scenarios | Probability to reach maturity (%) | 13.29 | 22.06 |
| | Probability of a capital loss (%) | 5.22 | 8.42 |
| | Expected loss at maturity (% nominal) | 3.05 | 4.92 |
| | Expected life (years) | 1.03 | 1.89 |
| | Expected number of coupons | 0.87 | 0.78 |
| | Expected coupon paid (% nominal) | 3.01 | 6.02 |

Table 18: Non-call period of one year, all other parameters being those of the base case.

The paths that were called at the first three dates in the base case now have to wait one year, and some of them fall in the meantime: the probability to reach maturity rises from 13.29% to 22.06% and the probability of a loss from 5.22% to 8.42%. Yet the fair coupon falls from 10.54% to 8.94%: since the coupon is a snowball, a note called after one year pays at least $c$, while most notes of the base case were called after three months with $c/4$. The coupon paid is twice as large on average (6.02% instead of 3.01%).

### 5.8.11 Coupon type

Finally, the coupon can be fixed instead of snowball: a call at $t_k$ pays $c\,(t_k - t_{k-1}) = c/4$, whatever the call date.

| | | **Base case** | **Fixed coupon** |
|---|---|---|---|
| Contract | Coupon type | Snowball | Fixed |
| Pricing | Price (% nominal) | 97.80 | 96.16 |
| | 95% confidence interval (±) | 0.11 | 0.10 |
| | Fair coupon (% p.a.) | 10.54 | 23.94 |
| Scenarios | Probability to reach maturity (%) | 13.29 | 13.29 |
| | Probability of a capital loss (%) | 5.22 | 5.22 |
| | Expected loss at maturity (% nominal) | 3.05 | 3.05 |
| | Expected life (years) | 1.03 | 1.03 |
| | Expected number of coupons | 0.87 | 0.87 |
| | Expected coupon paid (% nominal) | 3.01 | 1.30 |

Table 19: Fixed coupon: a call at $t_k$ pays $c\,(t_k - t_{k-1})$ instead of $c\,t_k$.

The scenarios are the same, but the coupon does not accumulate anymore: a note called after two years pays $c/4$ instead of $2c$. The coupon paid falls from 3.01% to 1.30% of the nominal, and the fair coupon per annum jumps to 23.94%. Quoted per annum, a fixed-coupon Athena looks very attractive, but it pays its coupon only once.

## 5.9 All the forms of the Athena

Table 20 gathers all the variants of this section, sorted by fair coupon, and Figure 13 draws their fair coupons.

| Variant | Price at 6% | Fair coupon (% p.a.) | P(maturity) (%) | P(loss) (%) | E[loss] (% nom.) | E[life] (y) | E[coupon] (% nom.) |
|---|---|---|---|---|---|---|---|
| Guaranteed 100% | 100.57 | 4.83 | 13.29 | 0.00 | 0.00 | 1.03 | 3.01 |
| Guaranteed 90% | 100.09 | 5.81 | 13.29 | 5.22 | 0.52 | 1.03 | 3.01 |
| Airbag | 99.11 | 7.83 | 13.29 | 5.22 | 1.61 | 1.03 | 3.01 |
| Step-down 1% | 98.81 | 8.24 | 8.12 | 4.44 | 2.62 | 0.87 | 3.29 |
| Duo 90% / 50% | 99.37 | 8.86 | 3.13 | 1.70 | 1.01 | 0.42 | 1.33 |
| Non-call 1 year | 97.15 | 8.94 | 22.06 | 8.42 | 4.92 | 1.89 | 6.02 |
| Look back min 1m | 98.75 | 9.08 | 9.03 | 3.57 | 2.09 | 0.78 | 2.51 |
| Annual observation | 96.61 | 9.41 | 25.70 | 9.06 | 5.27 | 2.06 | 6.22 |
| Semi-annual observation | 97.22 | 9.93 | 18.61 | 6.97 | 4.08 | 1.48 | 4.39 |
| **Base case** | 97.80 | 10.54 | 13.29 | 5.22 | 3.05 | 1.03 | 3.01 |
| Range accrual 80-120% | 97.68 | 10.99 | 13.29 | 5.22 | 3.05 | 1.03 | 2.88 |
| Monthly observation | 98.62 | 11.11 | 7.77 | 3.13 | 1.83 | 0.59 | 1.66 |
| Buffer 10% | 97.37 | 11.42 | 13.29 | 11.81 | 3.50 | 1.03 | 3.01 |
| Daily observation | 99.65 | 11.74 | 1.80 | 0.78 | 0.46 | 0.13 | 0.38 |
| American protection (daily) | 96.92 | 12.31 | 13.23 | 8.98 | 4.06 | 1.03 | 3.01 |
| American protection | 96.83 | 12.53 | 13.29 | 9.22 | 4.11 | 1.03 | 3.01 |
| Autocall 110% | 94.33 | 12.69 | 37.56 | 10.48 | 6.04 | 2.39 | 5.33 |
| Protection 80% | 96.60 | 13.00 | 13.29 | 9.57 | 4.34 | 1.03 | 3.01 |
| Fixed coupon | 96.16 | 23.94 | 13.29 | 5.22 | 3.05 | 1.03 | 1.30 |

Table 20: All the Athena variants of this section, sorted by fair coupon (Heston, 50,000 paths, coupon 6% p.a.). Each variant changes one feature of the base case.

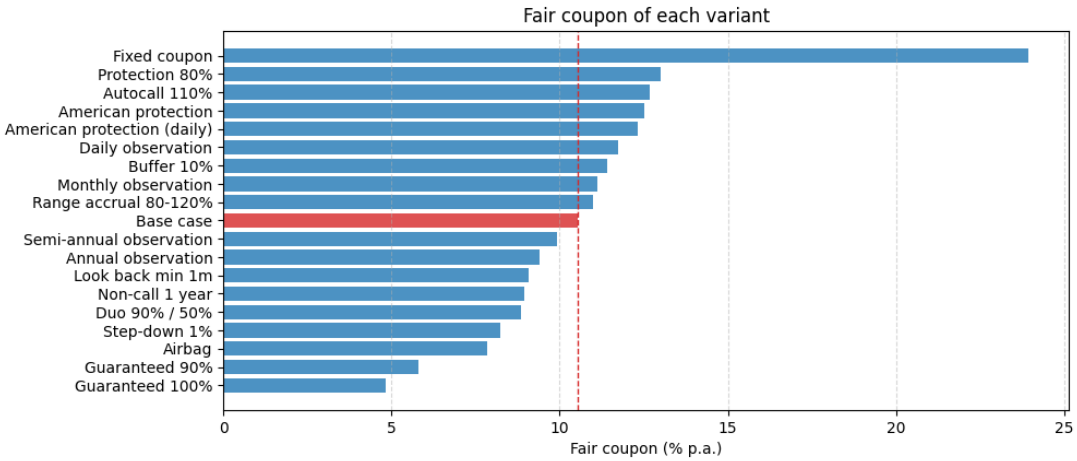



Figure 13: Fair coupon of each variant (the base case in red).

**Reading the table.** The investor is short a down-and-in put whose premium is paid back through the coupons; the fair coupon per annum is roughly this premium divided by the expected life of the note.

- *Capital protection.* The guarantee (4.83% at 100%, 5.81% at 90%) and the airbag (7.83%) reduce the loss without changing the scenarios; the American protection (12.53%), a higher barrier (13.00% at 80%) and the buffer (11.42%) make the loss more likely.
- *Call mechanism.* The features that make the call easier at a given coupon (step-down 8.24%, duo 8.86%, look back 9.08%) are more valuable for the investor; a higher autocall barrier is the opposite (12.69% at 110%).

- *Timing of the coupon.* Because the coupon is a snowball, everything that shortens the life of the note raises the coupon rate needed: more frequent observations (from 9.41% annual to 11.74% daily) and, conversely, a non-call period lowers it (8.94%). A fixed coupon, paid once, needs 23.94% a year.

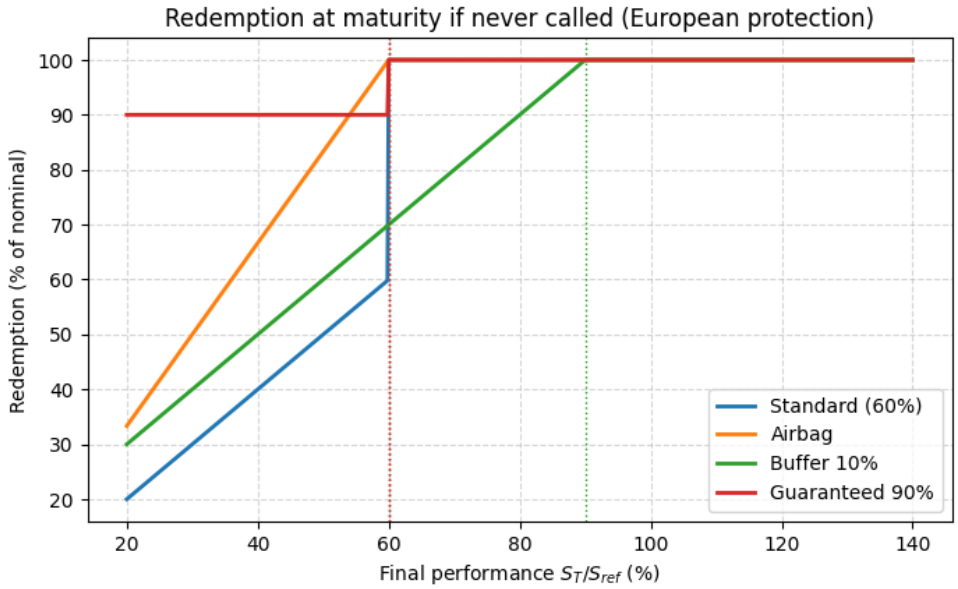



Figure 14: Redemption at maturity of a note never called: standard, airbag, buffer and capital guaranteed at 90%.

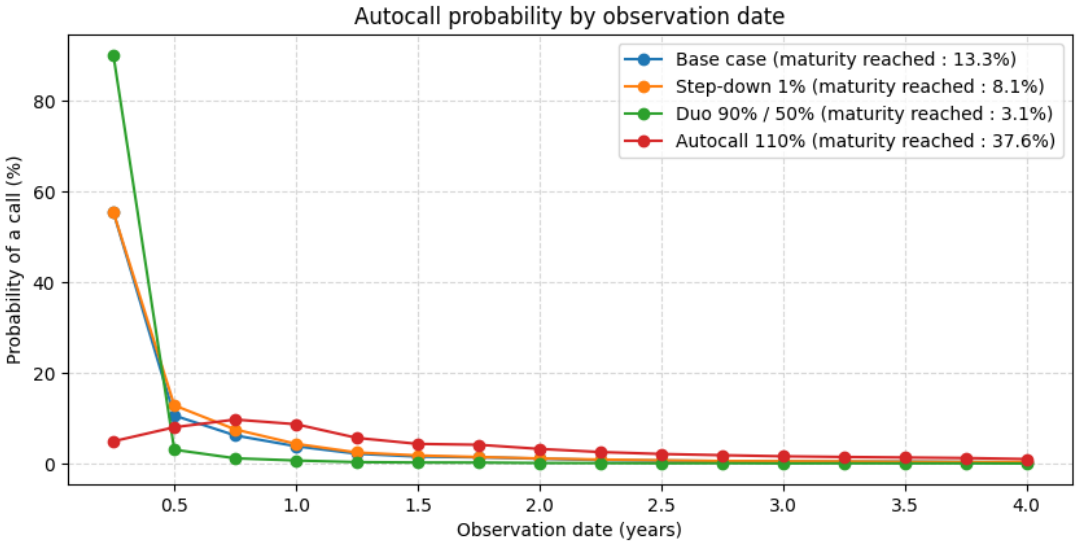



Figure 15: Probability of a call at each observation date: base case, step-down, duo and autocall barrier at 110%.

## 5.10 Greeks

Table 21 gives the Greeks of five Athena variants (nominal 100, coupon 6%). Gamma and Volga are not significant at this number of paths (base case: Gamma $-8.6 \times 10^{-6} \pm 5.1 \times 10^{-6}$, Volga $-0.022 \pm 0.028$); the other Greeks of the base case have relative standard errors between 4% and 35% (Delta $0.00470 \pm 0.00020$, Vega $-0.315 \pm 0.013$, Rho $-0.210 \pm 0.054$, Dividend $-0.689 \pm 0.054$, Theta $0.0048 \pm 0.0017$).

| **Athena** | **Price** | **Delta** (per point) | **Delta 1%** (per 1% move) | **Vega** (per vol pt) | **Vanna** ($\times 10^4$) | **Rho** (per 1%) | **Dividend** (per 1%) | **Theta** (per day) |
|---|---|---|---|---|---|---|---|---|
| Base case | 97.80 | 0.00470 | 0.378 | −0.315 | 2.45 | −0.210 | −0.689 | 0.0048 |
| American protection | 96.83 | 0.00640 | 0.515 | −0.375 | 2.04 | −0.058 | −0.801 | 0.0037 |
| Step-down 1% | 98.81 | 0.00225 | 0.181 | −0.299 | 2.86 | −0.421 | −0.354 | 0.0040 |
| Airbag | 99.11 | 0.00260 | 0.209 | −0.205 | 1.70 | −0.486 | −0.465 | 0.0024 |
| Guaranteed 90% | 100.09 | 0.00114 | 0.091 | −0.066 | 0.69 | −0.683 | −0.307 | 0.0019 |

Table 21: Greeks of Athena variants (nominal 100, coupon 6%).

The note is long the index (a 1% rise is worth 0.38 on 100) and short volatility: one vol point costs 0.32, the short put dominating the long digital-like call features. It is also short dividends: one point of carry lowers every forward and costs 0.69, twice the vega. It loses with higher rates (discounting of the coupons and of the nominal) and earns about 0.005 a day (1.7 a year) as the put decays. The American protection increases the delta and the vega of the note, since the knock-in can happen at any time, and brings the rho close to zero: higher rates raise the forwards and make the barrier less likely to be touched, which offsets the discounting. The step-down, the airbag and the capital guarantee reduce every risk but the rate risk, the guaranteed redemption behaving as a zero-coupon bond (rho −0.683). Figure 16 displays the table with 95% confidence intervals.

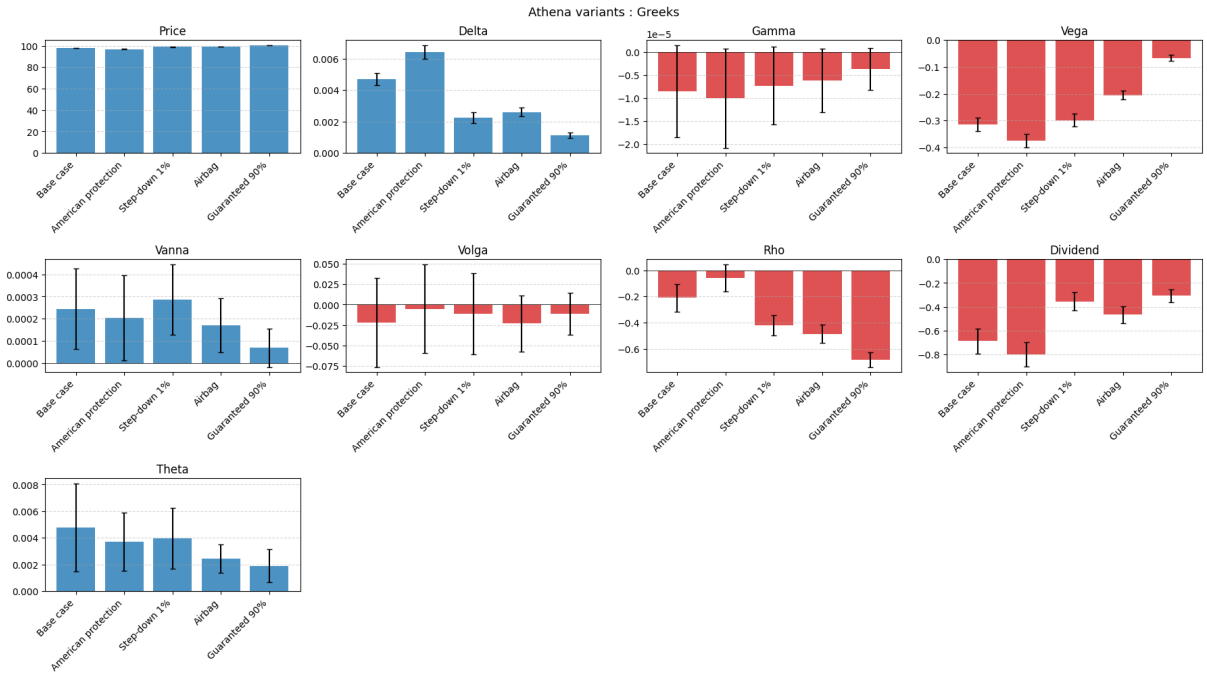



Figure 16: Greeks of the Athena variants with 95% confidence intervals.

**Sensitivities to the Heston parameters.** Per unit of parameter, the base case has $\partial V/\partial v_0 = -61.7 \pm 14.2$, $\partial V/\partial\theta = -18.6 \pm 9.0$, $\partial V/\partial\xi = 2.9 \pm 1.2$, while $\partial V/\partial\kappa = 0.11 \pm 0.26$

and $\partial V/\partial\rho = -1.1 \pm 0.8$ are not significant. The note loses with the variance level (short put), mostly through the initial variance $v_0$ since most notes are called within a year, and gains slightly with the vol-of-vol $\xi$. The insensitivity to $\rho$ is the one found in the skew scenarios above.

# 6 Implementation

## 6.1 Code

- `Products/Athena/athena.py`: the self-contained class `Athena` (contract, `path_payoff`, `fair_coupon`, `statistics`, and `features` / `report` for one table with the terms, the pricing and the scenarios);
- `Equity_Models/monte_carlo.py`: `MonteCarloPricer`, which simulates the paths and calls `path_payoff` for products with several cash flows;
- `Equity_Models/Heston.py`, `Equity_Models/forward_curve.py`: calibrated model and forward curve;
- `Equity_Models/greeks.py`: Greeks with common random numbers, including the Dividend Greek;
- `Products/market.py`: CAC40 market data and calibrated Heston model (`cac40_heston`);
- `Products/notes_analysis.py`: comparison of variants and plots;
- notebooks `Products/Athena/athena_pricer.ipynb` and `Products/Athena/athena_synthetic_pricing.ipynb` (every number of Section 5);
- `Reports/Athena/make_figures.py`: extracts the figures of this report from the executed notebook.

```
from market import cac40_heston
from athena import Athena
from Equity_Models.monte_carlo import MonteCarloPricer
from Equity_Models.greeks import Greeks

mkt = cac40_heston()
pricer = MonteCarloPricer(n_paths=50_000, steps_per_year=52, seed=42)
note = Athena(4.0, mkt.S0, 0.06, coupon_type="snowball", frequency=0.25,
              autocall_barrier=1.0, protection_barrier=0.6, protection_type="
                  european")
print(pricer.price(mkt.heston, note))              # price and standard error
print(note.fair_coupon(pricer, mkt.heston))        # coupon p.a. at par
print(note.report(pricer, mkt.heston))             # terms, pricing and scenario
    statistics
print(Greeks(pricer).compute(mkt.heston, note, Greeks.ORDER))
```

## 6.2 Cost

With $N$ paths and $M$ steps, the simulation costs $\mathcal{O}(NM)$ in time and stores $S$ and $v$, i.e. $16(M+1)N$ bytes: 167 MB for $N = 50{,}000$ and the weekly grid of 4 years, 807 MB on a daily grid. Following the note through its $n$ observation dates costs $\mathcal{O}(nN)$, the American continuous protection $\mathcal{O}(NM)$. Measured times (one CPU core): about 0.3 s for a valuation including the simulation, and a few hundredths of a second for the price, the fair coupon and the statistics of a variant on cached paths. The whole notebook `athena_synthetic_pricing.ipynb` (calibration, 19 variants, 6 market sweeps, the Greeks of five notes and a spot profile) runs in about 1.5 minutes.

# 7 Limitations and Next Steps

- *Single underlying.* Most autocalls sold today are worst-of on two to four indices or stocks; they need a multi-asset model and a correlation calibration.
- *Model.* The value of an autocall depends on the forward skew and on the joint dynamics of spot and volatility between observation dates; a pure Heston model with constant parameters is not the market standard, which is a local stochastic volatility model calibrated to the whole surface [6].
- *Fair coupon.* It prices the note at par without issuer margin; the credit spread of the issuer is only studied as a parallel discounting spread, without default modelling. A real issue price is below par by the margin.
- *Discretisation.* Euler full truncation with the Feller condition violated ($2\kappa\theta/\xi^2 = 0.49$), weekly grid with the variance frozen over each week for the Brownian bridge.
- *Greeks.* Gamma and Volga are not significant with 50,000 paths: the payoff jumps at the autocall and protection barriers.
- *Dividends.* Continuous dividend yield bootstrapped from the forwards; real notes are exposed to discrete dividends and to dividend risk, which the Dividend Greek only measures for a parallel shift.

# 8 Conclusion

A single parametrisation covers the Athena with all its usual features, and a Monte Carlo pricer under a calibrated Heston model recovers three exact limit cases within one standard error. On the CAC40 of 12 February 2025, the base 4-year quarterly Athena has a fair coupon of 10.54% a year. Its value is driven by the down-and-in put sold by the investor and by the short expected life of the note (1.03 years): everything that makes the loss more likely raises the coupon (protection barrier, American protection, buffer, volatility, dividends), everything that removes the loss lowers it (airbag, capital guarantee), and, because the coupon is a snowball, the features that shorten the life of the note (more frequent observations) raise the coupon rate needed while a non-call period lowers it. The rates and the issuer spread raise the coupon through the funding benefit of the issuer, and the skew has almost no effect on the price, its effects on the put and on the call offsetting each other. The note is short volatility and short dividends, the dividend exposure being twice the vega; the hedge of the issuer is dominated by the delta, which triples when the index falls towards the protection barrier.

# A Notation

| Symbol | Meaning |
|---|---|
| $N$, $S_{\text{ref}}$, $T$ | Nominal, initial fixing, maturity |
| $t_k$, $\delta_k = t_k - t_{k-1}$ | Observation dates and accrual periods |
| $P_k = S_{t_k}/S_{\text{ref}}$ | Performance at $t_k$ |
| $B_k^{\text{call}}$, $B^{\text{prot}}$ | Autocall and protection barriers (fractions of $S_{\text{ref}}$) |
| $c$, $c^\star$ | Coupon rate per annum, fair coupon |
| $K_{\text{put}}$, $\phi$, $b$ | Strike of the short put, capital floor, buffer |
| $\tau$ | Call date |
| $\mathbf{1}_{\text{KI}}$ | Knock-in event of the protection |

# References


[1] E. Benguedda, *Calibration of the Heston Stochastic Volatility Model: Theoretical Framework, Numerical Methods, and Limitations*, 2026.

[2] E. Benguedda, *Phoenix Autocallable Notes on the CAC40 under a Calibrated Heston Model: Fair Coupons, Scenarios and Greeks*, 2026.

[3] E. Benguedda, *Monte Carlo Pricing and Greeks of Single Barrier Options under a Calibrated Heston Model*, 2026.

[4] S. L. Heston, A closed-form solution for options with stochastic volatility with applications to bond and currency options, *Review of Financial Studies*, 6(2):327–343, 1993.

[5] M. Bouzoubaa and A. Osseiran, *Exotic Options and Hybrids: A Guide to Structuring, Pricing and Trading*, Wiley, 2010.

[6] L. Bergomi, *Stochastic Volatility Modeling*, Chapman and Hall/CRC, 2016.

[7] P. Glasserman, *Monte Carlo Methods in Financial Engineering*, Springer, 2004.

[8] D. R. Beaglehole, P. H. Dybvig and G. Zhou, Going to extremes: correcting simulation bias in exotic option valuation, *Financial Analysts Journal*, 53(1):62–68, 1997.

[9] E. Reiner and M. Rubinstein, Breaking down the barriers, *Risk*, 4(8):28–35, 1991.

[10] L. Andersen, Simple and efficient simulation of the Heston stochastic volatility model, *Journal of Computational Finance*, 11(3):1–42, 2008.